Stripe Payment Risk & Revenue Analytics

End-to-End Data Analytics Project Documentation

Python	Power BI	DAX	Power Query	PostgreSQL	Stripe REST API

Project Objective

To analyze real-world payment transactions and uncover revenue patterns, fee burden, and fraud risk across geographies, card types, and product categories using a complete end-to-end analytics pipeline.

Data Source

Data was collected using **Stripe's Test Mode REST API**, which mirrors the exact structure of production payment data. 3,200 transactions were extracted across 8 countries, 3 currencies, and 5 product categories. While transaction values are simulated, the entire pipeline follows the same workflow as real production data.

Metric	Value
Total Transactions	3,200
Countries	8 (US, GB, DE, FR, CA, AU, IN, SG)
Currencies	3 (USD, EUR, GBP)
Product Categories	5 (SaaS, E-commerce, Consulting, Subscription, Marketplace)
Card Brands	4 (Visa, Mastercard, Amex, Discover)
Tables Extracted	2 (Charges + Balance Transactions)

Project Workflow

Step 1 Python + API	Step 2 Power Query	Step 3 Star Schema	Step 4 DAX	Step 5 SQL	Step 6 Dashboard

Step 1 — Data Extraction (Python + Stripe API)

- Used Python and Stripe's REST API to programmatically generate and extract 3,200 test transactions
- Extracted two datasets: Charges (transaction details) and Balance Transactions (fee and net revenue)
- Saved raw data as CSV files for downstream processing

Step 2 — Data Cleaning (Power Query)

- Removed 13 redundant/null columns (failure codes, receipt emails, seller messages, etc.)

- Fixed data types — currency to decimal, timestamps to datetime, risk score to integer
- Standardized text casing across all categorical fields using Proper and Upper case functions
- Split datetime column into separate Date and Time components
- Kept raw tables in model view for interview reference and data lineage documentation

Step 3 — Star Schema Data Modeling (Power BI)

Built a proper Star Schema with 1 central Fact table and 4 Dimension tables:

Table	Type	Key Columns
Fact_Transactions	Fact	Charge Id, Amount, Amount USD, Risk Score, Time, Fee, Net + 4 FKs
Dim_Country	Dimension	Country Id, Customer Country, Full Country Name
Dim_Card	Dimension	Card Id, Card Brand, Card Funding
Dim_Currency	Dimension	Currency Id, Currency
Dim_Product	Dimension	Product Id, Product Category, Description

All relationships are One-to-Many (1:*) with Single filter direction from Dimension to Fact table. An Amount USD calculated column was added to normalize USD/EUR/GBP into a single currency using exchange rates.

Step 4 — DAX Measures

Folder	Measures
Revenue Measures	Total Revenue, Total Fees, Avg Transaction Value, Avg Fee Per Transaction
Risk Measures	Avg Risk Score, High Risk %, High Risk Count, High Risk Revenue
Transaction Measures	Transaction Count, High Value Count, Above Avg Risk Count
Profitability Measures	Net Revenue %, Revenue After Fees
Revenue Intelligence	Revenue Share %, Country Revenue Rank, % of Total Revenue, Low Risk Revenue %

Advanced DAX functions used: RANKX for revenue ranking, ALLSELECTED for dynamic share calculations, nested CALCULATE + FILTER for risk segmentation, DIVIDE for safe division, and RELATED for cross-table lookups.

Step 5 — SQL Analysis (PostgreSQL)

Loaded cleaned data into PostgreSQL database **stripe_analytics** and wrote 10 analytical queries:

#	Query Description	SQL Concepts Used
1	Rank countries by total revenue	RANK() window function
2	Transactions above category average	Correlated subquery
3	Cumulative revenue per country	Running total with ROWS UNBOUNDED
4	Top 3 transactions by country	ROW_NUMBER() + subquery
5	Fee efficiency by card brand	JOIN + aggregation
6	High risk + high value transactions	Window function + subquery
7	Product category performance	Revenue share window function
8	Above average risk card brands	HAVING + subquery
9	Revenue by currency with running total	SUM OVER + running total

10	High value vs normal by country	CASE WHEN aggregation

Step 6 — Power BI Dashboard

A single-page interactive dashboard with the following components:

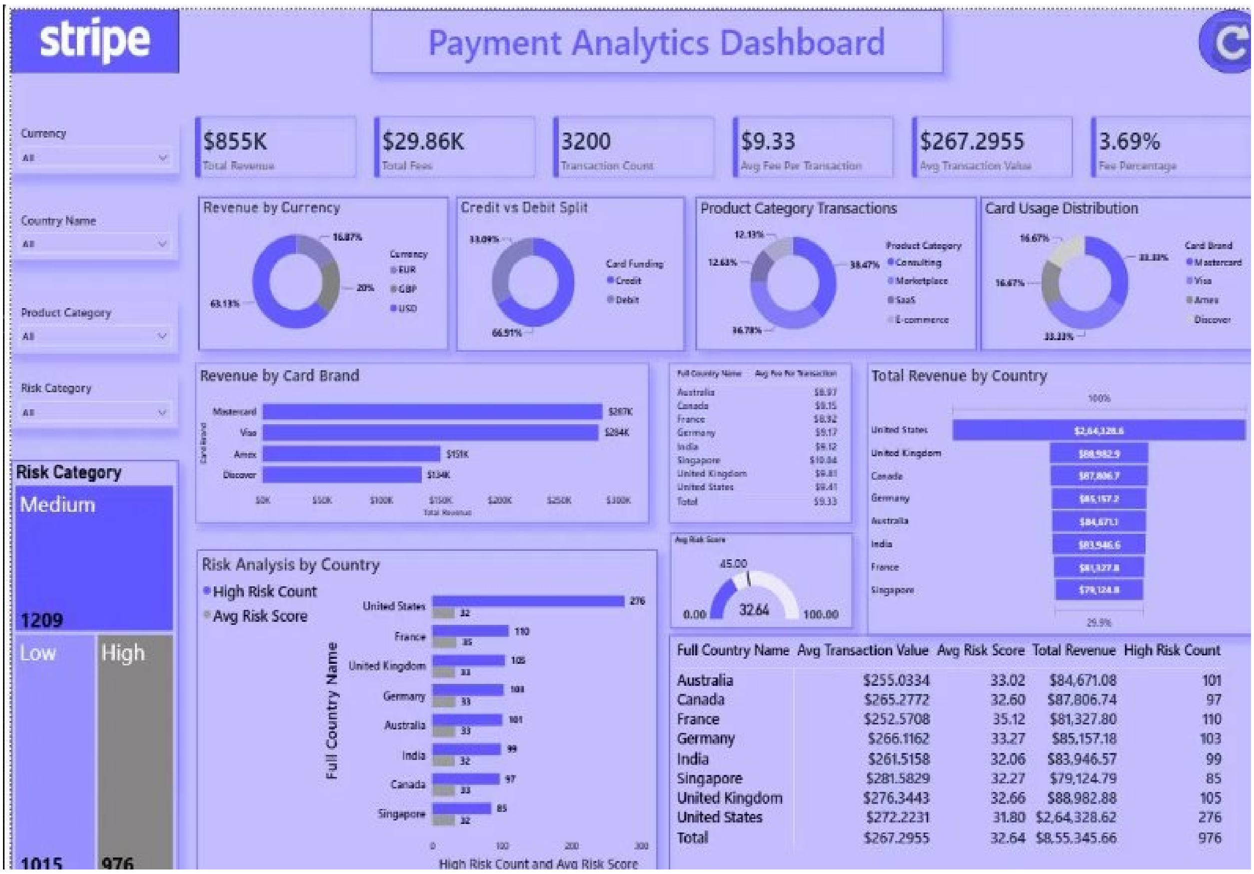
- **6 KPI Cards:** Total Revenue, Total Fees, Transaction Count, Avg Fee Per Transaction, Avg Transaction Value, Fee Percentage
- **Bar Charts:** Revenue by Country, Revenue by Card Brand, Risk Analysis by Country
- **Donut Charts:** Revenue by Currency, Credit vs Debit Split, Product Category Transactions, Card Usage Distribution
- **Gauge Chart:** Overall Avg Risk Score (0-100 scale with 45 threshold marker)
- **Matrix Table:** Country-level breakdown of Avg Transaction Value, Avg Risk Score, Total Revenue, High Risk Count
- **4 Dynamic Slicers:** Currency, Country, Product Category, Risk Category
- **Reset Button:** Clears all slicer selections with one click using bookmark action

Key Insights

#	Insight
1	United States drives 31% of total revenue — significantly ahead of all other countries
2	Credit cards account for 67% of all transactions vs 33% debit cards
3	30.5% of transactions fall into the high-risk category (risk score > 45)
4	Mastercard and Visa together contribute over 67% of total revenue
5	Average risk scores are consistent across countries (~32-35) — risk is not geography-driven
6	USD accounts for 63% of revenue, EUR 20%, and GBP 17%
7	Singapore has the highest avg transaction value ($281) despite lowest total revenue

Power BI Dashboard

Interactive single-page dashboard built in Power BI Desktop featuring KPI cards, bar charts, donut charts, gauge, matrix table, dynamic slicers, and a reset button.



Note on Data: This project uses Stripe's test mode, which generates simulated transaction data with the same structure as production data. The focus is demonstrating the complete analytics pipeline — from API extraction to dashboard — rather than deriving business insights from real transaction data.

Author: Tanmay Tiwari | KIIT Bhubaneswar, B.Tech CSE 2023–2027 | github.com/tanmay218